

Introduction

1 Data

- a) cross-sectional
- b) time series
- c) panel data

2 Variables

- a) continuous
- b) discrete
- c) qualitative
- d) quantitative

e) mix

Example

What are the determinants of household medical expenditures?

Discrete variables:

→ quantitative

→ qualitative

• ordered → e.g. customer satisfaction

• unordered e.g. color
makes, brands

strongly
dissatisfied
dissatisfied
neutral
satisfied
very satisfied

← basic
panel
models

truncated
limited variables

$y \in [a, b)$ $y \in [0, +\infty)$

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Example

What are the determinants of household medical expenditures?

- number of people
- number of diseases
- average age in the household
- number of children
- number of persons of age 65 or older
- income
- whether HH has medical insurance
- costs
-

LDV
 $y \in [0, +\infty)$

Plan of the course

The exact semielasticity
time always

$$[\exp(\beta) - 1] \cdot 100\%$$

$$[\exp(-1.1) - 1] \cdot 100\%$$

Let's consider the following model for military expenses among countries

$$\ln(ARMY_i) = 1.284 + 0.005 \ln(GDP_i) - 0.01 \ln(POPULATION_i)$$

$$- 1.10 \text{ noSEA}_i + \varepsilon_i$$

(6)

where ARMY denotes military budget in millions of USD, GDP stands for Gross Domestic Product in millions of USD, and noSEA is a dummy variable equal to 1 if a country has no direct access to international waters

$\beta \cdot 100\%$ approximation

$$|\beta| < 0.1$$

$$|\beta| < 0.5$$

$$\text{noSEA} = \begin{cases} 1 & \text{no access landlocked} \\ 0 & \text{otherwise} \end{cases}$$

$$\ln GDP \uparrow 1\% \Rightarrow E[\ln(ARMY)] \uparrow 0.005\%$$

$$\ln \text{Population} \uparrow 1\% \Rightarrow E[\ln(ARMY)] \downarrow 0.01\%$$

$$\text{noSEA}_{i=1} - \text{noSEA}_{i=0} \quad E[\ln(ARMY) | \text{noSEA}=1] - E[\ln(ARMY) | \text{noSEA}=0]$$

$$-1.1 \cdot 100\%$$

FOMC interest rate in the USA

$$\approx -110\% \text{ (not makes no sense)}$$

$$r = 2\%$$

$$\text{next month: } r' = 2\% + 1\% = 3\%$$

$$\text{the following month: } \underline{50\%} \text{ percent } r'' = 3\% + \underline{50\%} \cdot 3\% = 3\% + 1.5\% = 4.5\%$$

bare point

percents vs percentage points

↑ the difference is crucial